

ASSIGNMENT SET 8

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Subject: Cyber Law and Ethics

Code: OECIT801B

Q1/ Discuss the importance of cybersecurity awareness in preventing cybercrime.

- Cybersecurity awareness refers to educating users about cyber threats and safe practices while using digital systems and systems and the internet.
- It reduces the chances of cyber attacks caused by human error or lack of knowledge.
- Educates users about safe online practices like creating strong passwords and using two factor authentication.
- Helps people recognize suspicious mails.
- Encourages safe browsing and responsible use of digital devices.
- In organisations, trained employees act as the first line of defence against cyber attacks.
- Protects personal, financial, and organizational data from cybercriminals.
- Overall it reduces vulnerabilities and strengthens digital security.

Q2 Explain social engineering attacks with examples.

- Social engineering attacks involve manipulating people to reveal confidential
- These attacks exploit human emotions such as trust, fear, curiosity or urgency.
- Attackers trick user instead of exploiting technical system weaknesses

Examples

- Phishing: Fake emails or messages pretending to be from banks or trusted organizations asking for login details.
- Pretexting: An attacker pretending to be an authority figure such as IT support to obtain sensitive information.
- Baiting: Offering free software, downloads or gifts that actually contain malware.
- Tailgating: An unauthorized person follows an authorized employee to gain access to restricted areas.

Q3 Describe security threats in mobile operating systems.

- Malicious applications (Malware): Apps that steal personal information or damage the device.

- Phishing attacks: Fake SMS messages, emails, or apps that tricks users into revealing passwords or banking details
- Unsecured Wi-Fi networks: Attackers can intercept data transmitted over public Wi-Fi
- Operating system vulnerabilities: Outdated software may contain security weaknesses
- Data Theft: Personal data can be accessed if a device is lost or stolen without proper protection
- Spyware and tracking software: Can monitor user activities and location without permission

Q4 Explain the functioning of botnets in cyber attacks.

- A botnet is a network of infected computers or devices controlled by a cybercriminal
- The infected devices are called bots or zombies
- Devices become infected through malware, malicious downloads, or infected websites
- Bots connect to a Command and Control (C&C) server, which is controlled by the attacker
- The botmaster sends instructions to all infected devices simultaneously.

Uses of Botnets

- Launching Distributed Denial of Service (DDoS) attacks to overload servers

- Sending large amounts of spam emails
- Spreading malware to other devices
- Stealing sensitive information from devices victims.

Q5 Discuss the role of cyber forensic investigation in cyber law enforcement.

- Cyber forensic investigation involves collecting and analyzing digital evidence from electronic devices
- Helps identify the sources and methods of cyber crimes
- Investigators examine data from computers, mobile devices, servers, and networks
- Important evidence includes ~~is preserved~~ ~~per~~ ~~pro~~ emails, logs, files, browsing history and digital transactions
- Ensures that digital evidence is preserved properly without alteration
- Helps law enforcement agencies investigate crimes such as hacking, fraud, identity theft and data breaches
- Digital evidence collected through cyber forensics can be presented in court
- Supports the enforcement of cyber laws and helps bring ~~cyber~~ cybercriminals to justice.